

STOR 155.010 Lecture 11

Bayes' Theorem, Random Variables

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Overview

1 Bayes' Theorem

2 Random Variables

Bayes' Theorem: Two Outcomes

Let A, A^c partition Ω .

- **Law of Total Probability:**

$$\mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c) = \mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c).$$

- **Bayes' Theorem:** Given event B has occurred:

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)}.$$

- **Proof:**

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B|A)\mathbb{P}(A) + \mathbb{P}(B|A^c)\mathbb{P}(A^c)}$$

Example: Two Outcomes

- 5% of all emails are spam.
- Spam emails contain the word “free” 60% of the time.
- Non-spam emails contain “free” only 5% of the time.
- An email arrives containing the word “free.”
- What's the probability this email is actually spam?

Solution: Two Outcomes

- Let S = “spam,” F = “contains ‘free’.”
- Given: $\mathbb{P}(S) = 0.05$, $\mathbb{P}(F|S) = 0.6$, $\mathbb{P}(F|S^c) = 0.05$.
- Law of Total Probability:

$$\mathbb{P}(F) = (0.05)(0.6) + (0.95)(0.05) = 0.03 + 0.0475 = 0.0775$$

- Bayes' Theorem:

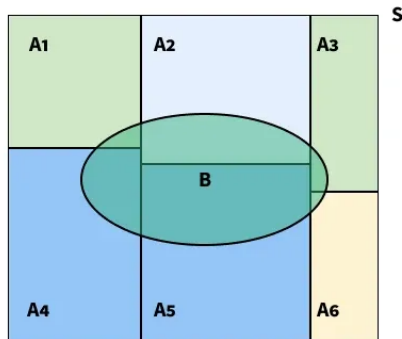
$$\mathbb{P}(S|F) = \frac{(0.6)(0.05)}{0.0775} = \frac{0.03}{0.0775} \approx 0.387$$

- Only about a 38.7% chance the email is spam, even with the trigger word present.

Law of Total Probability

Let $A_1, \dots, A_k$ **partition** Ω : disjoint ($A_i \cap A_j = \emptyset$ for all $i \neq j$), and covering all outcomes ($\bigcup_{i=1}^k A_i = \Omega$). For any event $B \subseteq \Omega$,

$$\mathbb{P}(B) = \sum_{i=1}^k \mathbb{P}(B|A_i) \mathbb{P}(A_i)$$



Example: Law of Total Probability

- Factory has 3 machines A, B, C, producing items in proportions:

$$\mathbb{P}(A) = 0.5, \quad \mathbb{P}(B) = 0.3, \quad \mathbb{P}(C) = 0.2$$

- A, B, C partition Ω : every item from exactly one machine, proportions sum to 1.
- Let D = “item is defective.” Defect rates per machine:

$$\mathbb{P}(D|A) = 0.01, \quad \mathbb{P}(D|B) = 0.02, \quad \mathbb{P}(D|C) = 0.03$$

- Law of Total Probability:

$$\mathbb{P}(D) = \mathbb{P}(D|A)\mathbb{P}(A) + \mathbb{P}(D|B)\mathbb{P}(B) + \mathbb{P}(D|C)\mathbb{P}(C) = 0.017$$

- Overall, 1.7% of items are defective, weighted by each machine's output share.

Bayes' Theorem: k Outcomes

- Same idea as the two-outcome case, now with a full partition $A_1, \dots, A_k$.
- Denominator: just the Law of Total Probability, applied to B .
- Given event B has occurred, probability of A_i :

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(B|A_i)\mathbb{P}(A_i)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A_i)\mathbb{P}(B|A_i)}{\mathbb{P}(A_1)\mathbb{P}(B|A_1) + \dots + \mathbb{P}(A_k)\mathbb{P}(B|A_k)}$$

Example: Three Outcomes

- Back to the factory: machines A, B, C produce 50%, 30%, 20% of items.
- Defect rates: 1%, 2%, 3% respectively.
- A randomly selected item turns out to be defective.
- What's the probability it came from machine C?

Solution: Three Outcomes

- Let D = “defective.” Already know, from Law of Total Probability:

$$\mathbb{P}(D) = (0.5)(0.01) + (0.3)(0.02) + (0.2)(0.03) = 0.017$$

- Bayes' Theorem, applied to machine C:

$$\mathbb{P}(C|D) = \frac{\mathbb{P}(C) \mathbb{P}(D|C)}{\mathbb{P}(D)} = \frac{(0.2)(0.03)}{0.017} = \frac{0.006}{0.017} \approx 0.353$$

- About a 35.3% chance the defective item came from machine C, despite C making only 20% of items, since its defect rate is highest.

Example in Medicine

- A disease affects 2% of the population.
- A diagnostic test correctly identifies a sick person 95% of the time (true positive rate).
- The test incorrectly flags a healthy person as positive 3% of the time (false positive rate).
- A randomly selected person tests positive.
- What is the probability they actually have the disease?

Solution

- Let D = “has disease,” T = “tests positive.”
- Given: $\mathbb{P}(D) = 0.02$, $\mathbb{P}(T|D) = 0.95$, $\mathbb{P}(T|D^c) = 0.03$.
- By the Law of Total Probability:

$$\mathbb{P}(T) = \mathbb{P}(T|D)\mathbb{P}(D) + \mathbb{P}(T|D^c)\mathbb{P}(D^c)$$

$$\mathbb{P}(T) = (0.95)(0.02) + (0.03)(0.98) = 0.019 + 0.0294 = 0.0484$$

- By Bayes' Theorem:

$$\mathbb{P}(D|T) = \frac{\mathbb{P}(T|D)\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{(0.95)(0.02)}{0.0484} = \frac{0.019}{0.0484} \approx 0.392$$

- Even with a positive test, there's only about a 39.2% chance of actually having the disease.

Overview

- 1 Bayes' Theorem
- 2 Random Variables

Introduction to Random Variables

- We are often interested in numerical outcomes of random processes (for example, number of heads in 10 tosses of a coin).
- A **random variable (RV)** is a function that assigns a real number to each outcome in a sample space.
- Turns every possible outcome of a random experiment into a numerical value.
- Denoted by $X : \Omega \rightarrow \mathbb{R}$. For example, pick a person at random and measure their height.

Discrete RV:

- Takes finite or countably infinitely many values.
- Examples: number of heads in 10 coin tosses, number of defective items in a batch, number of students coming late to class today, etc.

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Continuous RV:

- Can take on any value in $\mathbb{R}$ (or an interval). Uncountably many possible values.
- Examples: time taken to run a marathon, amount of rain in a city over a year, etc.

Discrete Random Variables

A discrete RV can be summarized by its elementary outcomes and their associated probabilities.

Probability Mass Function (pmf): $f_X(x) = \mathbb{P}(X = x)$.

Dice sum	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Rules:

- 1 Each probability must be between 0 and 1.
- 2 The probabilities must sum to 1.

Expectation, Variance, and Standard Deviation

Suppose X is a discrete RV taking values in $\{x_1, \dots, x_K\} \subset \mathbb{R}$.

- **Expected value/mean:** weighted average of its possible outcomes.

$$\mu = \mathbb{E}[X] = \sum_{i=1}^K x_i \mathbb{P}(X = x_i).$$

- **Variance:** measures variability in realized values

$$\sigma^2 = \text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_{i=1}^K (x_i - \mathbb{E}[X])^2 \cdot \mathbb{P}(X = x_i).$$

- **Standard Deviation:**

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}.$$